# LEMMA ALGEBRAIZATION PRINCIPAL

THE STACKS PROJECT AUTHORS

*Editorial note.* All mathematical statement and proof text is reproduced verbatim from the author's chapter algebraization.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \external-document list are omitted because the assembler supplies the class.

*Editorial note.* Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

*Editorial note.* The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-lemma-algebraization-principal.

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*Allowed standard prerequisites (editorial).*

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

*Source reference.* Original reference 0EHC: sources/algebraization.tex, source lines 4983–4983. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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*Source reference.* Original lemma 0EHA: sources/cohomology.tex, source lines 9262–9262. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original reference 0BK3: sources/local-cohomology.tex, source lines 1728–1728. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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*Source reference.* Original lemma 0DXG: sources/cohomology.tex, source lines 9365–9365. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0DX1: sources/algebraization.tex, source lines 664–664. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0AAK: sources/dualizing.tex, source lines 3510–3510. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original proposition 08Z9: sources/dualizing.tex, source lines 968–968. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0A7Z: sources/dualizing.tex, source lines 3281–3281. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original reference 0DWZ: sources/local-cohomology.tex, source lines 1968–1968. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *In Situation 0EHC let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Assume*

(1) *$A$ is local and $\mathfrak{a} = \mathfrak{m}$ is the maximal ideal,*
(2) *$A$ has a dualizing complex,*
(3) *$I = (f)$ is a principal ideal for a nonzerodivisor $f \in \mathfrak{m}$,*
(4) *$\mathcal{F}_n$ is a finite locally free $\mathcal{O}_U/f^n\mathcal{O}_U$-module,*
(5) *if $\mathfrak{p} \in V(f) \setminus \{\mathfrak{m}\}$, then $depth((A/f)_\mathfrak{p}) + \dim(A/\mathfrak{p}) > 1$, and*
(6) *if $\mathfrak{p} \notin V(f)$ and $V(\mathfrak{p}) \cap V(f) \neq \{\mathfrak{m}\}$, then $depth(A_\mathfrak{p}) + \dim(A/\mathfrak{p}) > 3$.*

*Then $(\mathcal{F}_n)$ extends canonically to $X$. In particular, if $A$ is complete, then $(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_U$-module.*

*Proof.* We will prove this by verifying hypotheses (a), (b), and (c) of Lemma 0EHH.

Since $\mathcal{F}_n$ is locally free over $\mathcal{O}_U/f^n\mathcal{O}_U$ we see that we have short exact sequences $0 \to \mathcal{F}_n \to \mathcal{F}_{n+1} \to \mathcal{F}_1 \to 0$ for all $n$. Thus condition (b) holds by Cohomology, Lemma 0EHA.

By induction on $n$ and the short exact sequences $0 \to A/f^n \to A/f^{n+1} \to A/f \to 0$ we see that the associated primes of $A/f^nA$ agree with the associated primes of $A/fA$. Since the associated points of $\mathcal{F}_n$ correspond to the associated primes of $A/f^nA$ not equal to $\mathfrak{m}$ by assumption (3), we conclude that $M_n = H^0(U, \mathcal{F}_n)$ is a

finite $A$-module by (5) and Local Cohomology, Proposition 0BK3. Thus hypothesis (c) holds.

To finish the proof it suffices to show that there exists an $n > 1$ such that the image of

$$H^1(U, \mathcal{F}_n) \longrightarrow H^1(U, \mathcal{F}_1)$$

has finite length as an $A$-module. Namely, this will imply hypothesis (a) by Cohomology, Lemma 0DXG. The image is independent of $n$ for $n$ large enough by Lemma 0DX1. Let $\omega_A^\bullet$ be a normalized dualizing complex for $A$. By the local duality theorem and Matlis duality (Dualizing Complexes, Lemma 0AAK and Proposition 08Z9) our claim is equivalent to: the image of

$$\mathrm{Ext}_A^{-2}(M_1, \omega_A^\bullet) \to \mathrm{Ext}_A^{-2}(M_n, \omega_A^\bullet)$$

has finite length for $n \gg 1$. The modules in question are finite $A$-modules supported at $V(f)$. Thus it suffices to show that this map is zero after localization at a prime $\mathfrak{q}$ containing $f$ and different from $\mathfrak{m}$. Let $\omega_{A_\mathfrak{q}}^\bullet$ be a normalized dualizing complex on $A_\mathfrak{q}$ and recall that $\omega_{A_\mathfrak{q}}^\bullet = (\omega_A^\bullet)_\mathfrak{q}[\dim(A/\mathfrak{q})]$ by Dualizing Complexes, Lemma 0A7Z. Using the local structure of $\mathcal{F}_n$ given in (4) we find that it suffices to show the vanishing of

$$\mathrm{Ext}_{A_\mathfrak{q}}^{-2+\dim(A/\mathfrak{q})}(A_\mathfrak{q}/f, \omega_{A_\mathfrak{q}}^\bullet) \to \mathrm{Ext}_{A_\mathfrak{q}}^{-2+\dim(A/\mathfrak{q})}(A_\mathfrak{q}/f^n, \omega_{A_\mathfrak{q}}^\bullet)$$

for $n$ large enough. If $\dim(A/\mathfrak{q}) > 3$, then this is immediate from Local Cohomology, Lemma 0DWZ. For the other cases we will use the long exact sequence

$$\ldots \xrightarrow{f^n} H^{-1}(\omega_{A_\mathfrak{q}}^\bullet) \to \mathrm{Ext}_{A_\mathfrak{q}}^{-1}(A_\mathfrak{q}/f^n, \omega_{A_\mathfrak{q}}^\bullet) \to H^0(\omega_{A_\mathfrak{q}}^\bullet) \xrightarrow{f^n} H^0(\omega_{A_\mathfrak{q}}^\bullet) \to \mathrm{Ext}_{A_\mathfrak{q}}^0(A_\mathfrak{q}/f^n, \omega_{A_\mathfrak{q}}^\bullet) \to 0$$

If $\dim(A/\mathfrak{q}) = 2$, then $H^0(\omega_{A_\mathfrak{q}}^\bullet) = 0$ because $\mathrm{depth}(A_\mathfrak{q}) \geq 1$ as $f$ is a nonzerodivisor. Thus the long exact sequence shows the condition is that

$$f^{n-1} : H^{-1}(\omega_{A_\mathfrak{q}}^\bullet)/f \to H^{-1}(\omega_{A_\mathfrak{q}}^\bullet)/f^n$$

is zero. Now $H^{-1}(\omega_\mathfrak{q}^\bullet)$ is a finite module supported in the primes $\mathfrak{p} \subset A_\mathfrak{q}$ such that $\mathrm{depth}(A_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) \leq 1$. Since $\dim((A/\mathfrak{p})_\mathfrak{q}) = \dim(A/\mathfrak{p}) - 2$ condition (6) tells us these primes are contained in $V(f)$. Thus the desired vanishing for $n$ large enough. Finally, if $\dim(A/\mathfrak{q}) = 1$, then condition (5) combined with the fact that $f$ is a nonzerodivisor insures that $A_\mathfrak{q}$ has depth at least 2. Hence $H^0(\omega_{A_\mathfrak{q}}^\bullet) = H^{-1}(\omega_{A_\mathfrak{q}}^\bullet) = 0$ and the long exact sequence shows the claim is equivalent to the vanishing of

$$f^{n-1} : H^{-2}(\omega_{A_\mathfrak{q}}^\bullet)/f \to H^{-2}(\omega_{A_\mathfrak{q}}^\bullet)/f^n$$

Now $H^{-2}(\omega_\mathfrak{q}^\bullet)$ is a finite module supported in the primes $\mathfrak{p} \subset A_\mathfrak{q}$ such that $\mathrm{depth}(A_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) \leq 2$. By condition (6) all of these primes are contained in $V(f)$. Thus the desired vanishing for $n$ large enough. $\square$